UNIT 3 REVIEW: EQUATIONS OF LINES AND PLANES

Equation of Lines

2. Find a vector equation of the line
   a. that passes through the points $(3, 9)$ and $(-4, 2)$
   b. that passes through the point $(-5, -3)$ and is parallel to the line $\vec{r} = (4, 0) + t(0, 5)$
   c. that is perpendicular to the line $2x - 5y - 6 = 0$ and passes through the point $(0, -3)$

3. Find parametric equations of the line
   a. that passes through $(-9, 8)$ with slope $-\frac{2}{3}$
   b. that passes through $(3, -2)$ and is perpendicular to the line $\vec{r} = (4, -1) + t(3, 2)$
   c. through the points $(4, 0)$ and $(0, -2)$

4. Find a vector equation of the line
   a. that passes through the points $(2, 0, -3)$ and $(-3, 2, -2)$
   b. that has an x-intercept of -7 and a z-intercept of 4
   c. that is parallel to $\frac{x-5}{4} = \frac{y+2}{-2} = \frac{z+6}{5}$ and passes through the point $(0, 6, 0)$

5. Find parametric equations of the line
   a. that is parallel to the line $\frac{x+1}{-3} = \frac{y+2}{-2} = z + 3$ and passes through the origin
   b. that passes through the point $(6, -4, 5)$ and is parallel to the y-axis

6. Find the Cartesian equation of the line
   a. that passes through the point $(-1, -2)$ and is parallel to the line $3x - 4y + 5 = 0$
   b. that passes through the point $(-7, 3)$ and is perpendicular to the line $x = 2 + t, y = -3 + 2t$
   c. that passes through the origin and is perpendicular to the line $x + 4y + 1 = 0$

7. a. Find the parametric equations of the line l that passes through the point $A(6, 4, 0)$ and is parallel to the line passing through $B(-2, 0, 4)$ and $C(3, -2, 1)$.
   b. If $(-4, m, n)$ is a point on l, find m and n.

8. Determine if the following pairs of lines are parallel and distinct, coincident, perpendicular, or none of these.
   a. $\vec{r} = (2, 3) + t(-3, 1)$ and $\vec{r} = (-1, 4) + u(6, -2)$
   b. $x = 1 + 2t, y = -3 - t$ and $x = u, y = \frac{1}{3} + 2u$
   c. $\frac{x-1}{2} = \frac{y+4}{1}, z = 1$ and $x = 4t, y = 1 + 2t, z = 6$
   d. $(x, y, z) = (1, 7, 2) + t(-1, -1, 1)$ and $(x, y, z) = (-3, 0, 1) + u(2, -2, -2)$

10. In the xy-plane,
    a. find the Cartesian equation of the line $\vec{r} = (2, 3) + t(-1, 5)$
    b. find a vector equation of the line $5x - 2y + 10 = 0$
    c. find a vector equation of the line $y = \frac{3}{4}x + \frac{1}{2}$

12. Find the direction cosines and the direction angles (to the nearest degree) of the direction vectors of the following lines.
    a. $\frac{x-3}{5} = \frac{y+6}{2} = \frac{z-1}{-1}$
    b. $x = 1 + 8t, y = 2 - t, z = 4 - 4t$
    c. $\vec{r} = (-7, 0, 0) + t(4, 1, 0)$

14. Find the shortest distance between
    a. the points $(2, 1, 3)$ and $(0, -4, 7)$
    b. the point $(3, 7)$ and the line $2x - 3y = 7$
    c. the point $(4, 0, 1)$ and the line $\vec{r} = (2, -2, 1) + t(1, 2, -1)$
    d. the point $(1, 3, 2)$ and the line $\frac{x-1}{-1} = \frac{y-3}{1} = \frac{z-7}{2}$

Review Exercise

2. a. $\vec{r} = (3, 9) + t(1, 1)$ **b.** $\vec{r} = (-5, -3) + t(1, 0)$ **c.** $\vec{r} = (0, -3) + t(2, -5)$ **3. a.** $x = -9 + 3t, y = 8 - 2t$ **b.** $x = 3 + 2s, y = -2 - 3s$ **c.** $x = 4 + 2t, y = t$ **4. a.** $\vec{r} = (2, 0, -3) + t(5, -2, -1)$ **b.** $\vec{r} = (-7, 0, 0) + t(7, 4, 0)$ **c.** $\vec{r} = (0, 6, 0) + t(4, -2, 5)$ **5. a.** $x = 3t, y = 2t, z = -t$ **b.** $x = 6, y = -4 + t, z = 5$ **c.** $x = t, y = -3t, z = -3 + 6t$ **6. a.** $3x - 4y - 5 = 0$ **b.** $x + 2y + 1 = 0$ **c.** $4x - y = 0$ **7. a.** $x = 6 + 5t, y = 4 - 2t, z = -3t$ **b.** $m = 8, n = -2$ **8. a.** coincident **b.** perpendicular **c.** parallel and distinct **d.** neither parallel nor perpendicular **9.** $\left(-3, \frac{11}{2}, 0\right)$, $(8, 0, 22)$, $(0, 4, 6)$ **10. a.** $5x + y - 13 = 0$ **b.** $\vec{r} = (0, 5) + t(2, 5)$ **c.** $\vec{r} = (2, 2) + t(4, 3)$ **11. a.** $(6, 0, 0)$, $(0, 8, 4)$ **b.** x intercept is 6 **12. a.** $\cos\alpha = \frac{5}{\sqrt{30}}$, $\cos\beta = \frac{2}{\sqrt{30}}$, $\cos\gamma = \frac{-1}{\sqrt{30}}$; $\alpha \cong 24°, \beta \cong 69°, \gamma \cong 101°$

b. $\cos\alpha = \frac{8}{9}$, $\cos\beta = \frac{-1}{9}$, $\cos\gamma = \frac{-4}{9}$; $\alpha \cong 27°, \beta \cong 96°, \gamma \cong 116°$ **c.** $\cos\alpha = \frac{4}{\sqrt{17}}$, $\cos\beta = \frac{1}{\sqrt{17}}$, $\cos\gamma = 0$; $\alpha \cong 14°, \beta \cong 76°, \gamma = 90°$ **13. a.** $(0, 0, 2)$ **14. a.** $3\sqrt{5}$ **b.** $\frac{22\sqrt{13}}{13}$ **c.** $\sqrt{2}$ **d.** $\frac{5\sqrt{3}}{3}$ **15.** $(4, -1, 5)$

Mistakes and notes about answers:

2b) $\vec{m} = (0,1)$ 4b) $\vec{m} = (7, 0, 4)$ 7b) n=6 8d) neither 12) the direction cosines should be rationalized (no radicals in denominators)

Equation of Planes

2. Find vector and parametric equations of the plane
   a. that passes through the point $(-1, -1, 2)$ and is parallel to the plane $\vec{r} = (2, -1, 0) + s(5, 4, 2) + t(0, 0, 1)$
   b. that passes through the points $(1, 1, 0)$ and $(-2, 0, 3)$ and is parallel to the y-axis

3. Find the scalar equation for the plane
   a. that passes through the point $(1, 7, 9)$ and has normal $\vec{n} = (1, 3, 5)$
   b. that passes through the points $(3, 2, 3)$, $(-4, 1, 2)$, and $(-1, 3, 2)$
   c. that passes through the point $(0, 0, 6)$ and is parallel to the plane $y + z = 5$
   d. that contains the point $(3, -3, 0)$ and the line $x = 2, y = 3 + t, z = -4 - 2t$
   e. that contains the line $\vec{r} = (2, 1, 7) + s(0, 1, 0)$ and is parallel to the line $\vec{r} = (3, 0, 4) + t(2, -1, 0)$
   f. that contains the points $(6, 1, 0)$ and $(3, 0, 2)$, and is parallel to the z-axis

4. For what value of k, if any, will the planes $3x + ky + z - 6 = 0$ and $6x + (1 - k)y + 2z - 9 = 0$ be
   a. parallel? b. perpendicular?

5. Find the scalar equation of the plane that contains the parallel and distinct lines $x = 1, \frac{y-3}{4} = \frac{z}{2}$ and $x = 5, \frac{y+5}{2} = \frac{z-3}{1}$.

6. Find a vector equation of the plane that contains the origin and the point $(2, -3, 2)$ and is perpendicular to the plane $x + 2y - z + 3 = 0$.

7. Find the scalar equation of the plane that passes through the point $(1, 2, 3)$ and is parallel to the vectors $6\hat{k}$ and $\hat{i} + 2\hat{j} - 3\hat{k}$.

8. A line that passes through the origin intersects a plane at the point $(1, -3, 2)$. If the line is perpendicular to the plane, find the scalar equation of the plane.

9. Find the scalar equation of the plane that contains the intersecting lines $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-1}{-1}$ and $\frac{x-1}{-1} = \frac{y-1}{5} = \frac{z-1}{4}$.

10. Explain why the point $(2, 21, 8)$ and the line $\vec{r} = (-4, -3, -1) + t(2, 8, 3)$ do not determine a plane.

11. Find the distance between
   a. the point $(7, 7, -7)$ and the plane $6y - z + 5 = 0$
   b. the point $(3, 2, 1)$ and the plane $3x+2y+z=10$
   c. the line $\vec{r} = (1, 3, 2) + t(1, 2, -1)$ and the plane $y + 2z = 5$
   d. the planes $x + 2y - 5z - 10 = 0$ and $2x + 4y - 10z - 17 = 0$

12. Find the distance from the point $(1, -2, -2)$ to the plane having an x-intercept of -1, a y-intercept of 2, and a z-intercept of 3.

Review Exercise

2. a. $\vec{r} = (-1, -1, 2) + s(5, 4, 2) + t(0, 0, 1)$; $x = -1 + 5s$, $y = -1 + 4s$, $z = 2 + 2s + t$
b. $\vec{r} = (1, 1, 0) + s(0, 1, 0) + t(3, 1, -3)$; $x = 1 + 3t$, $y = 1 + s + t$, $z = -3t$
c. $\vec{r} = (0, 0, 4) + s(2, -3, 0) + t(1, 0, 2)$; $x = 2s + t$, $y = -3s$, $z = 4 + 2t$
d. $\vec{r} = s(1, 1, 1) + t(3, 4, 5)$; $x = s + 3t$, $y = s + 4t$, $z = s + 5t$
e. $\vec{r} = (3, -1, 2) + s(4, 0, 1) + t(4, 0, 2)$ $x = 3 + 4s + 4t$, $y = -1$, $z = 2 + s + 2t$ **3. a.** $x + 3y + 5z - 67 = 0$
b. $2x - 3y - 11z + 33 = 0$ **c.** $y + z - 6 = 0$
d. $8x + 2y + z - 18 = 0$ **e.** $z - 7 = 0$ **f.** $x - 3y - 3 = 0$
4. a. $\frac{1}{3}$ **b.** $k = 5$ or $k = -4$ **5.** $7x + 2y - 4z - 13 = 0$
6. $\vec{r} = s(1, 2, -1) + t(2, -3, 2)$ **7.** $2x - y = 0$
8. $x - 3y + 2z - 14 = 0$ **9.** $17x - 7y + 13z - 23 = 0$
11. a. $\frac{54}{\sqrt{37}}$ **b.** $\frac{4}{\sqrt{14}}$ **c.** $\frac{2}{\sqrt{5}}$ **d.** $\frac{3}{2\sqrt{30}}$ **12.** $\frac{22}{7}$ **13.** $\left(\frac{4}{5}, \frac{-2}{5}, 1\right)$